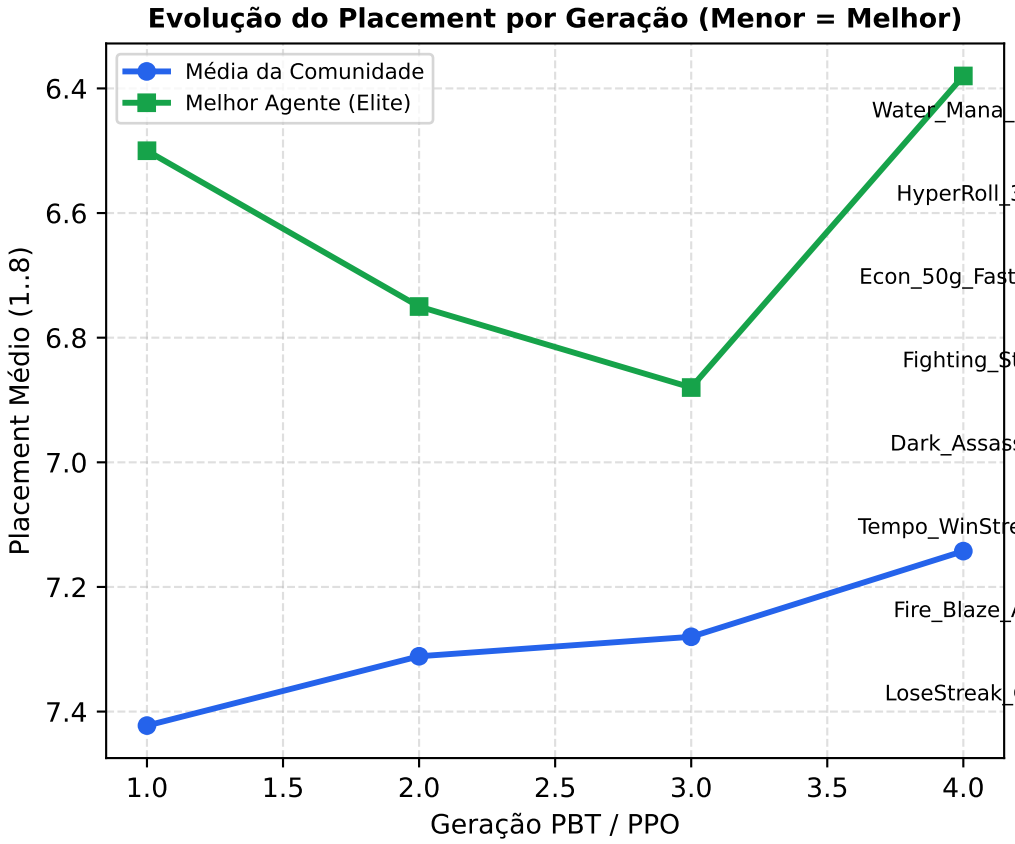
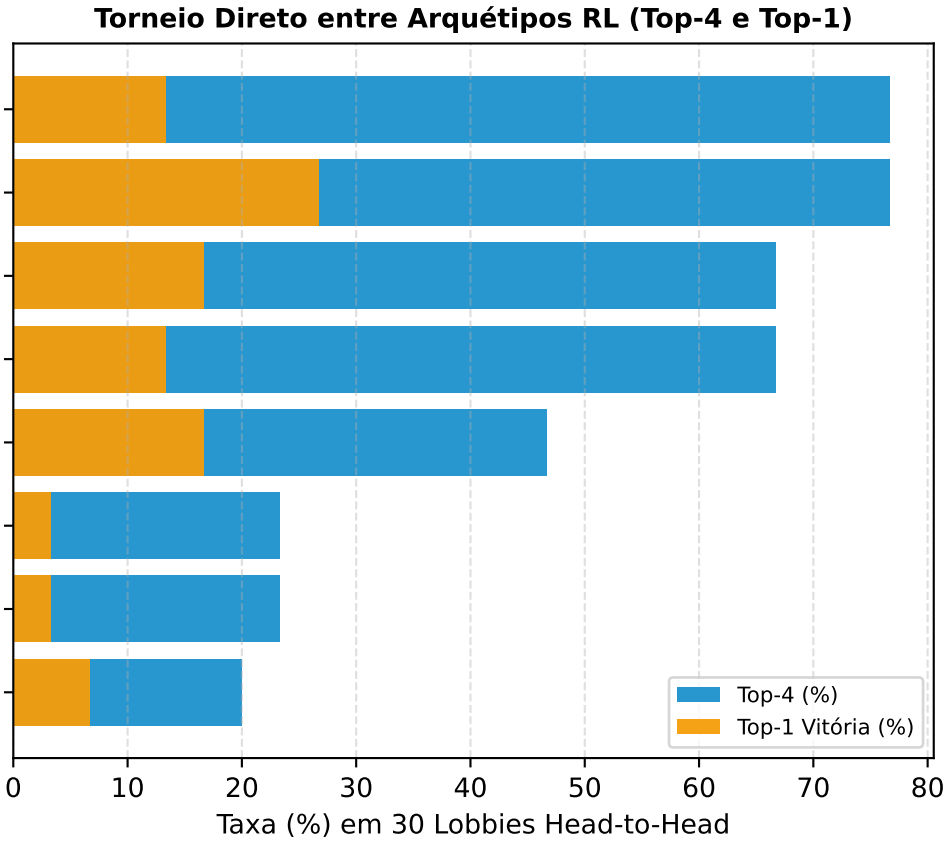
POKÉMON AUTO CHESS (PAC v6.11) — RELATÓRIO DA COMUNIDADE RL & HEURÍSTICAS VIPER

1. DADOS OFICIAIS DO REPOSITÓRIO (PAC v6.11):
   • Pokémon Jogáveis Carregados: 1077 | Sinergias Oficiais: 31 | Bots Comunitários (`botv2.json`): 726
   • Regras Econômicas Exatas: Reroll = 1g | Level Up = 4g (+4 XP) | Juros = min(5, floor(gold/10)) | Streak = min(5, streak) | +1g por vitória PvP.

2. MODELO SURROGATE DE COMBATE (`SetTransformerCombatSurrogate` em PyTorch `nn.MultiheadAttention`):
   • Acurácia de Validação na Previsão de Vitória: 90.7% | Erro Médio Absoluto (Unidades Sobreviventes): 1.57 unidades (amostras=2000)

3. ARQUITETURA DA COMUNIDADE DE AGENTES RL:
   • Políticas `RewardConditionedMaskedActorCritic` (FiLM Conditioning + Invalid Action Masking) treinadas com Maskable PPO + GAE.
   • Seleção Evolucionária por Population-Based Training (PBT) + Arquivo Qualidade-Diversidade MAP-Elites + Destilação VIPER.



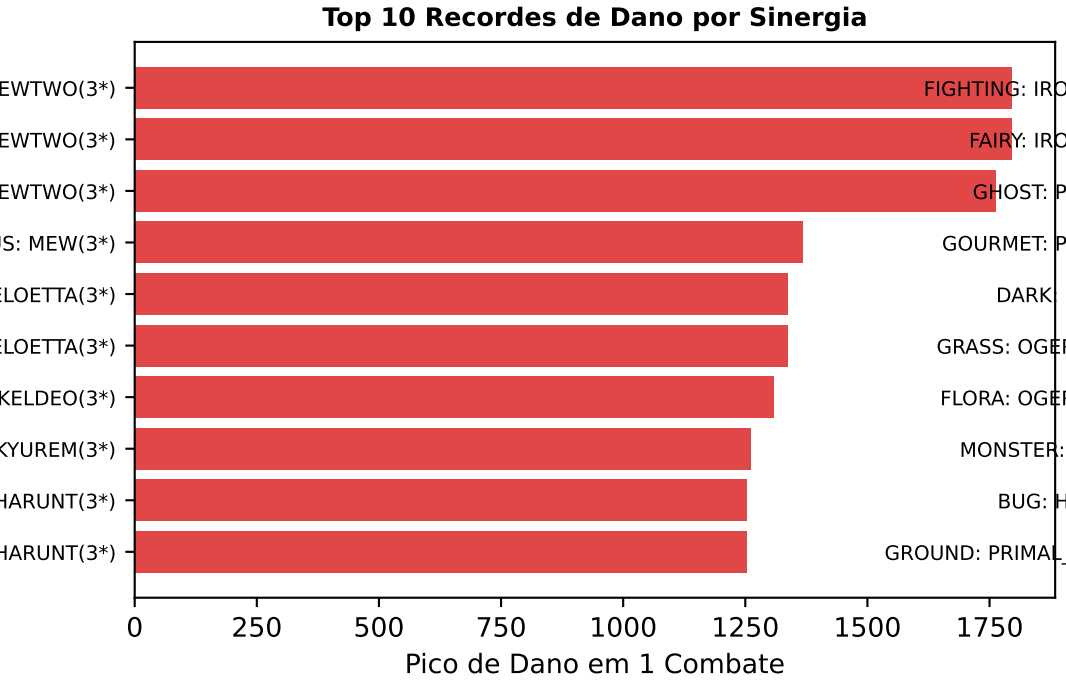


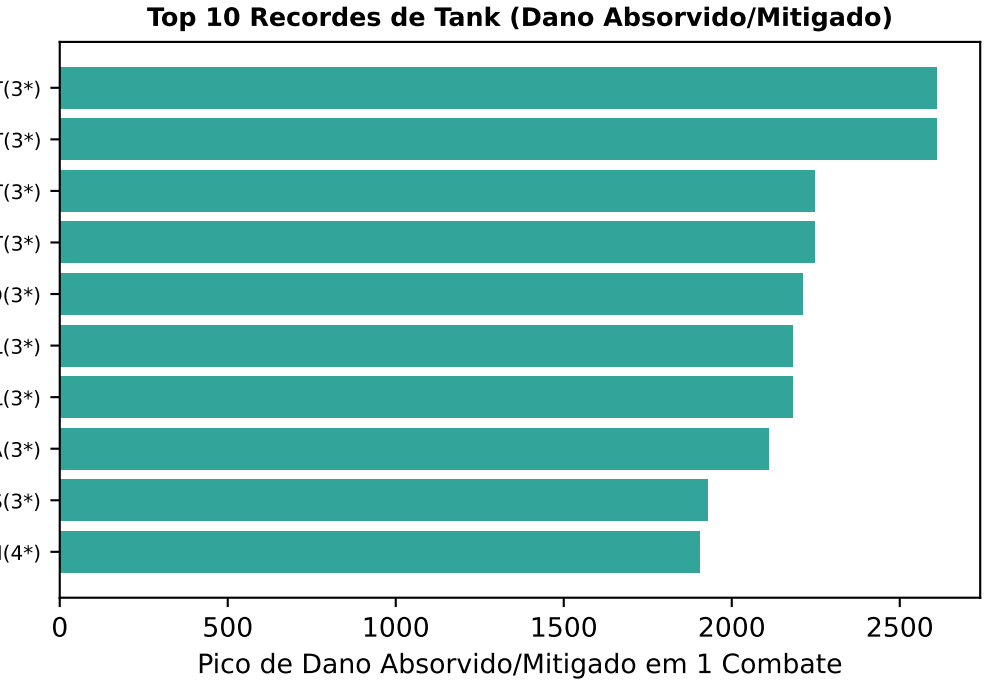
RECORDES DE DANO (MELHOR CARRY) E DE TANK (MELHOR FRONTLINE) POR COMPOSIÇÃO / SINERGIA

Sinergia	Recordista Dano (Carry	Recorde Dano	Dano Médio	2º Melhor Carry	Recordista Tank (Front	Recorde Tank	Tank Médio	2º Melhor Tank
MONSTER	MEWTWO(3*)	1794.6	657.0	GARCHOMP(3*) (max 1249	TORTERRA(3*)	2109.9	1165.5	PRIMAL_GROUDON(4*) (max 1906)
ARTIFICIAL	MEWTWO(3*)	1794.6	663.7	IRON_VALIANT(3*) (max 118	SILVALLY(3*)	1579.4	750.2	IRON_VALIANT(3*) (max 1638)
PSYCHIC	MEWTWO(3*)	1763.3	701.0	MEW(3*) (max 1368)	SOLGALEO(3*)	1262.6	977.6	GALLADE(3*) (max 1504)
AMORPHOUS	MEW(3*)	1367.8	697.7	NIHILEGO(3*) (max 1115)	MELMETAL(3*)	1281.5	886.6	EELEKTROSS(3*) (max 987)
SOUND	MELOETTA(3*)	1336.5	532.5	PIROUETTE_MELOETTA(3*) (max	MARACTUS(3*)	1655.4	869.4	PIROUETTE_MELOETTA(3*) (max 131
NORMAL	MELOETTA(3*)	1336.5	527.4	PIROUETTE_MELOETTA(3*) (max	REGIGIGAS(3*)	1584.1	794.2	CHANSEY(2*) (max 1532)
AQUATIC	KELDEO(3*)	1309.1	429.8	SHADOW_LUGIA(3*) (max 11	WALREIN(3*)	1303.6	765.1	LUGIA(3*) (max 1120)
ICE	KYUREM(3*)	1261.6	520.6	TORNADUS(3*) (max 1026	REGICE(3*)	1463.7	940.1	ABOMASNOW(2*) (max 1675)
POISON	PECHARUNT(3*)	1254.1	661.5	NIHILEGO(3*) (max 1134)	VENUSAUR(3*)	1460.6	608.2	GLIMMORA(2*) (max 1484)
GOURMET	PECHARUNT(3*)	1254.1	548.7	HYDRAPPLE(4*) (max 1186	PECHARUNT(3*)	2247.9	515.6	CHANSEY(2*) (max 1532)
GHOST	PECHARUNT(3*)	1254.1	540.2	ORIGIN_GIRATINA(3*) (max 8	PECHARUNT(3*)	2247.9	505.2	SPECTRIER(3*) (max 1441)
DRAGON	GARCHOMP(3*)	1249.0	581.8	MEGA_RAYQUAZA(4*) (max 1	REGIDRAGO(3*)	1541.3	1051.0	DIALGA(3*) (max 1459)
HUMAN	GALLADE(3*)	1247.2	530.8	INFERNAPE(3*) (max 1124	LILIGANT(2*)	1675.0	598.6	REGIROCK(3*) (max 1347)
FIGHTING	GALLADE(3*)	1247.2	530.8	KELDEO(3*) (max 1309)	IRON_VALIANT(3*)	2608.9	697.6	BRELOOM(2*) (max 2099)
ELECTRIC	THUNDURUS(3*)	1220.9	721.3	ZEKROM(3*) (max 1038)	ZAPDOS(3*)	1077.8	520.7	IRON_HANDS(3*) (max 1018)
FLYING	MEGA_RAYQUAZA(4*)	1200.7	595.8	HO_OH(3*) (max 1166)	SHAYMIN_SKY(4*)	1843.3	580.3	BEAUTIFLY(3*) (max 1251)
DARK	SHADOW_LUGIA(3*)	1195.4	472.2	HYDREIGON(3*) (max 990	GUZZLORD(3*)	2211.6	684.2	KROOKODILE(3*) (max 1265)
FIRE	HO_OH(3*)	1166.1	468.5	RESHIRAM(3*) (max 983)	PRIMAL_GROUDON(4*)	1469.5	1049.8	INFERNAPE(3*) (max 1489)
FAIRY	TINKATON(3*)	1161.2	574.0	IRON_VALIANT(3*) (max 118	IRON_VALIANT(3*)	2608.9	823.6	CHANSEY(2*) (max 1532)
STEEL	TINKATON(3*)	1161.2	574.0	AEGISLASH(3*) (max 977	STAKATAKA(3*)	1883.8	1072.7	DIALGA(3*) (max 1459)
ROCK	NIHILEGO(3*)	1115.5	465.0	SUDOWOODO(2*) (max 112	STAKATAKA(3*)	1883.8	1118.3	SUDOWOODO(2*) (max 1768)
LIGHT	ULTRA_NECROZMA(4*)	1106.3	590.3	HO_OH(3*) (max 1141)	LURANTIS(2*)	1682.4	911.9	SOLGALEO(3*) (max 1263)
FOSSIL	ROARING_MOON(3*)	1094.4	557.4	REGIROCK(3*) (max 759)	REGIDRAGO(3*)	1541.3	1239.5	REGICE(3*) (max 1464)
GROUND	ZYGARDE_50(3*)	1087.2	519.8	LANDORUS(3*) (max 879)	PRIMAL_GROUDON(4*)	1905.8	1250.7	TORTERRA(3*) (max 1704)

RECORDES DE DANO E DE TANK POR COMPOSIÇÃO (CONTINUAÇÃO — TODAS AS 31 SINERGIAS)

Sinergia	Recordista Dano (Carry	Recorde Dano	Dano Médio	2º Melhor Carry	Recordista Tank (Front	Recorde Tank	Tank Médio	2º Melhor Tank
WATER	PRIMAL_KYOGRE(4*)	1017.1	422.9	MANAPHY(3*) (max 917	PALKIA(3*)	1009.8	650.9	SAMUROTT(3*) (max 933)
FLORA	SHAYMIN_SKY(4*)	987.7	557.1	XERNEAS(3*) (max 1001	OGERPON_TEAL(3*)	2180.7	1633.0	MARACTUS(3*) (max 1655)
GRASS	SHAYMIN_SKY(4*)	987.7	557.1	VIRIZION(3*) (max 865)	OGERPON_TEAL(3*)	2180.7	1633.0	TORTERRA(3*) (max 2110)
FIELD	SERPERIOR(3*)	892.4	625.7	HEATMOR(3*) (max 881	SAMUROTT(3*)	1023.5	759.5	SUICUNE(3*) (max 1025)
BUG	GENESECT(3*)	847.3	508.5	BEAUTIFLY(3*) (max 673	HERACROSS(3*)	1929.3	645.5	BEAUTIFLY(3*) (max 1251)
WILD	PRIMEAPE(2*)	749.2	269.5	SUICUNE(3*) (max 667)	EMBOAR(3*)	1216.4	559.0	RAIKOU(3*) (max 1056)
BABY	CHINGLING(2*)	381.7	150.2	HAPPINY(1*) (max 140)	HAPPINY(1*)	616.9	484.0	CHINGLING(2*) (max 472)





HALL DA FAMA GLOBAL — TOP 15 POKÉMON RECORDISTAS DE DANO (DPS) E DE TANK NO PAC

Top 15 Pokémon com Maior Recorde Absoluto de Dano Causado

Pokemon	Rarity	Types	Matches	Max_Damage_Recor	Avg_Damage	Max_Tank_Record	Avg_Tanked	Avg_Spells_Cast	Avg_Kills
MEWTWO(3*)	LEGENDARY	PSYCHIC/MONSTER/ARTIFI	109	1794.6	637.8	1857.2	348.8	2.21	2.55
MEW(3*)	LEGENDARY	PSYCHIC/AMORPHOUS	32	1367.8	683.0	713.9	442.8	5.66	2.41
MELOETTA(3*)	LEGENDARY	NORMAL/SOUND	143	1336.5	523.0	1113.8	375.5	4.52	2.03
KELDEO(3*)	LEGENDARY	AQUATIC/FIGHTING	20	1309.1	429.8	829.2	543.5	2.9	1.35
KYUREM(3*)	LEGENDARY	DRAGON/ICE	89	1261.6	520.6	937.2	313.2	2.25	2.03
PECHARUNT(3*)	LEGENDARY	POISON/GHOST/GOURM	51	1254.1	540.2	2247.9	505.2	1.96	1.98
GARCHOMP(3*)	EPIC	DRAGON/GROUND/MONS	16	1249.0	581.8	1109.2	659.4	2.31	2.25
GALLADE(3*)	EPIC	PSYCHIC/FIGHTING/HUM	32	1247.2	530.8	1503.7	555.7	1.41	2.0
THUNDURUS(3*)	LEGENDARY	FLYING/ELECTRIC	12	1220.9	606.3	748.0	409.8	4.0	2.58
MEGA_RAYQUAZA(4*)	LEGENDARY	DRAGON/FLYING	69	1200.7	595.8	1297.7	461.8	1.81	2.49
SHADOW_LUGIA(3*)	LEGENDARY	AQUATIC/FLYING/DARK	50	1195.4	472.2	1000.2	492.7	2.64	2.02
HYDRAPPLE(4*)	UNIQUE	DRAGON/GOURMET/GRA	37	1186.3	306.1	627.4	119.6	1.68	1.3

Top 15 Pokémon com Maior Recorde Absoluto de Dano Tankado / Mitigado

Pokemon	Rarity	Types	Matches	Max_Damage_Recor	Avg_Damage	Max_Tank_Record	Avg_Tanked	Avg_Spells_Cast	Avg_Kills
IRON_VALIANT(3*)	LEGENDARY	FIGHTING/FAIRY/ARTIFIC	57	1185.8	543.3	2608.9	690.2	4.37	2.23
PECHARUNT(3*)	LEGENDARY	POISON/GHOST/GOURM	51	1254.1	540.2	2247.9	505.2	1.96	1.98
GUZZLORD(3*)	LEGENDARY	GOURMET/DARK/MONST	85	926.6	306.1	2211.6	673.3	1.88	1.15
OGERPON_TEAL(3*)	LEGENDARY	GRASS/FLORA	6	660.2	528.0	2180.7	1633.0	4.33	1.33
TORTERRA(3*)	RARE	GRASS/GROUND/MONST	24	602.7	279.7	2109.9	1036.0	2.38	0.71
BRELOOM(2*)	UNCOMMON	GRASS/FIGHTING	22	658.9	253.6	2098.6	813.3	2.5	0.82
HERACROSS(3*)	UNIQUE	BUG/FIGHTING	12	816.8	388.5	1929.3	645.5	2.25	1.75
SHAYMIN_SKY(4*)	LEGENDARY	GRASS/FLORA/FLYING	30	987.7	557.1	1925.4	587.5	2.93	2.23
PRIMAL_GROUDON(4*	LEGENDARY	GROUND/MONSTER/FIR	23	757.7	314.2	1905.8	1250.7	3.17	0.78
STAKATAKA(3*)	LEGENDARY	ROCK/STEEL	41	531.8	249.8	1883.8	983.2	2.41	0.95
MEWTWO(3*)	LEGENDARY	PSYCHIC/MONSTER/ARTIFI	109	1794.6	637.8	1857.2	348.8	2.21	2.55
SUDOWOODO(2*)	EPIC	ROCK/FLORA/MONSTEI	70	1124.0	218.9	1768.4	839.0	1.86	0.6

TORNEIO DIRETO ENTRE OS AGENTES RL (COM RECORDISTAS) & MATRIZ MAP-ELITES

Agent_Name	Archetype	Target_Synergy	Avg_Placement	Top4_Rate_%	Win_Rate_%	Avg_Bank_Gold	Avg_Final_Level	Record_DPS_Unit	Record_Tank_Unit
Water_Mana_Control_Spec	STANDARD_ECON_50G	WATER	3.33	76.7	13.3	79.2	9.0	PRIMAL_KYOGRE(4*) (1017	PALKIA(3*) (1010 tank)
HyperRoll_3Star_Collect	HYPER_ROLL	NORMAL	3.4	76.7	26.7	53.0	7.93	MELOETTA(3*) (1336 dps)	GIGIGIGAS(3*) (1584 tank)
Econ_50g_Fast8_Compou	STANDARD_ECON_50G	FLEX	3.53	66.7	16.7	56.3	8.3	MELOETTA(3*) (1336 dps)	GIGIGIGAS(3*) (1584 tank)
ighting_Steel_Bastion_Spe	STANDARD_ECON_50G	FIGHTING	3.8	66.7	13.3	12.2	8.37	GALLADE(3*) (1247 DPS)	IRON_VALIANT(3*) (2609 tank)
Dark_Assassin_Crit_Speci	HYPER_ROLL	DARK	4.67	46.7	16.7	101.4	6.93	SHADOW_LUGIA(3*) (1165	BUZZLORD(3*) (2212 tank)
Tempo_WinStreak_Snowb	TEMPO_WIN_STREAK	FLEX	5.53	23.3	3.3	42.6	7.77	MELOETTA(3*) (1336 dps)	GIGIGIGAS(3*) (1584 tank)
Fire_Blaze_Aggro_Specia	TEMPO_WIN_STREAK	FIRE	5.8	23.3	3.3	42.7	7.67	HO_OH(3*) (1166 DPS)	PRIMAL_GROUDON(4*) (1470 tan
LoseStreak_Cashout_Pi	LOSE_STREAK_CASHOU	ARTIFICIAL	5.93	20.0	6.7	50.1	6.67	MEWTWO(3*) (1795 dps)	SILVALLY(3*) (1579 tank)

Synergy_Niche	Econ_Archetype	Best_Agent	Avg_Placement	Top4_Rate_%	Win_Rate_%	Signature_Board
FIGHTING	STANDARD_ECON_50G	Fighting_Steel_Bastion_Specialist	6.38	12.5	DOLLIV(2*),	GOLBAT(2*), QUILLADIN(2*), MACHOP(1*), OSHAWO
DARK	HYPER_ROLL	Dark_Assassin_Crit_Specialist	6.38	12.5	LARVITAR	(1*), WIGLETT(1*), FLORAGATO(2*), DEINO(1*), ABS
FIRE	STANDARD_ECON_50G	Econ_50g_Fast8_Compounder	6.5	0.0	SWINUB(	1*), METAPOD(2*), PIPLUP(1*), KOFFING(1*), DRILBU
NORMAL	HYPER_ROLL	HyperRoll_3Star_Collector	6.88	0.0	HERDIER(2*	), QUILLADIN(2*), SLIGOO(2*), GREAVARD(1*), MILTA
ARTIFICIAL	LOSE_STREAK_CASHOUT	LoseStreak_Cashout_Pivot	7.0	12.5	DOLLIV(2*),O	KLANG(2*), PRINPLUP(2*), ROTOM_FROST(3*), ELECTR
WATER	STANDARD_ECON_50G	Water_Mana_Control_Specialist	7.12	0.0	WARTORTLE(	2*), FENNEKIN(1*), FORRETRESS(2*), KAKUNA(2*), SE
FIRE	TEMPO_WIN_STREAK	Fire_Blaze_Aggro_Specialist	7.25	0.0	LITWICK(1*),C	HARMANDER(1*), MARILL(2*), TENTACOOL(1*), FENN
FIGHTING	TEMPO_WIN_STREAK	Tempo_WinStreak_Snowballer	7.25	0.0	TANGELA(	1*), BRAIXEN(2*), PIKACHU(2*), LICKILICKY(2*), AIPOM
WILD	STANDARD_ECON_50G	Econ_50g_Fast8_Compounder	7.38	0.0	ZUBAT(	1*), ZUBAT(1*), LILIGANT(2*), FEAROW(2*), LILEEP(1*
FAIRY	TEMPO_WIN_STREAK	Tempo_WinStreak_Snowballer	7.5	0.0	FLABEBE(	1*), FLAFFY(2*), GRIMER(1*), BRAIXEN(2*), JIGGLYPUF
MONSTER	TEMPO_WIN_STREAK	Tempo_WinStreak_Snowballer	7.75	0.0	PILOSWINE(	2*), ESPURR(1*), DEWPIDER(1*), AMBIPOM(2*), PIKA

DESTILAÇÃO DE POLÍTICAS NEURAIS EM REGRAS HUMANAS (VIPER / DECISION TREES)

■ Econ_50g_Fast8_Compounder [STANDARD_ECON_50G] — Fidelidade da Árvore vs Rede Neural PyTorch: 64.8%
  Variáveis Decisivas: gold (31.3%), stage_level (30.9%), player_hp (19.5%), shop_pair_matches (13.2%), free_board_slots (3.8%), bench_size (1.5%)
  • SE (stage_level > 8.50 E player_hp <= 43.50 E shop_pair_matches <= 0.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=74.1%, n=528)
  • SE (stage_level > 8.50 E player_hp > 43.50 E gold > 53.50) ENTÃO [COMPRAR_XP_SUBIR_NIVEL_4G] (confiança=58.3%, n=505)
  • SE (stage_level > 8.50 E player_hp > 43.50 E gold <= 53.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=58.5%, n=444)

■ Tempo_WinStreak_Snowballer [TEMPO_WIN_STREAK] — Fidelidade da Árvore vs Rede Neural PyTorch: 79.9%
  Variáveis Decisivas: player_hp (57.9%), bench_size (32.7%), gold (9.4%)
  • SE (player_hp <= 63.50 E bench_size > 5.50 E player_hp <= 41.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=99.0%, n=494)
  • SE (player_hp <= 63.50 E bench_size > 5.50 E player_hp > 41.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=87.9%, n=400)
  • SE (player_hp > 63.50 E bench_size > 5.50 E gold > 13.50) ENTÃO [COMPRAR_XP_SUBIR_NIVEL_4G] (confiança=50.5%, n=374)

■ HyperRoll_3Star_Collector [HYPER_ROLL] — Fidelidade da Árvore vs Rede Neural PyTorch: 79.7%
  Variáveis Decisivas: bench_size (43.3%), gold (37.3%), player_hp (15.6%), shop_synergy_matches (2.3%), num_2star_units (1.5%)
  • SE (bench_size > 5.50 E gold > 11.50 E player_hp > 60.50) ENTÃO [REROLL_LOJA_1G] (confiança=81.9%, n=1047)
  • SE (bench_size > 5.50 E gold <= 11.50 E num_2star_units > 1.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=78.7%, n=756)
  • SE (bench_size <= 5.50 E player_hp > 53.50 E gold > 0.50) ENTÃO [COMPRAR_UNIDADE_SINERGIA_OU_PAR] (confiança=97.3%, n=386)

■ LoseStreak_Cashout_Pivot [LOSE_STREAK_CASHOUT] — Fidelidade da Árvore vs Rede Neural PyTorch: 65.2%
  Variáveis Decisivas: bench_size (48.2%), player_hp (31.3%), shop_pair_matches (9.1%), shop_synergy_matches (6.1%), gold (5.4%)
  • SE (bench_size > 5.50 E player_hp > 29.00 E player_hp <= 64.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=70.5%, n=481)
  • SE (bench_size > 5.50 E player_hp <= 29.00 E shop_synergy_matches > 3.50) ENTÃO [REROLL_LOJA_1G] (confiança=58.0%, n=281)
  • SE (bench_size > 5.50 E player_hp > 29.00 E player_hp > 64.50) ENTÃO [COMPRAR_XP_SUBIR_NIVEL_4G] (confiança=28.3%, n=250)

■ Fire_Blaze_Aggro_Specialist [TEMPO_WIN_STREAK] — Fidelidade da Árvore vs Rede Neural PyTorch: 74.7%
  Variáveis Decisivas: bench_size (48.2%), player_hp (29.3%), gold (12.5%), shop_synergy_matches (10.0%)
  • SE (bench_size > 5.50 E player_hp <= 67.00 E shop_synergy_matches <= 3.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=90.7%, n=388)
  • SE (bench_size > 5.50 E player_hp > 67.00 E gold > 13.50) ENTÃO [COMPRAR_XP_SUBIR_NIVEL_4G] (confiança=54.1%, n=383)
  • SE (bench_size > 5.50 E player_hp > 67.00 E gold <= 13.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=58.7%, n=337)

■ Water_Mana_Control_Specialist [STANDARD_ECON_50G] — Fidelidade da Árvore vs Rede Neural PyTorch: 80.4%
  Variáveis Decisivas: bench_size (62.7%), shop_synergy_matches (13.7%), free_board_slots (8.3%), gold (8.0%), lose_streak (7.3%)
  • SE (bench_size > 5.50 E free_board_slots <= 0.50 E gold <= 53.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=79.8%, n=2078)
  • SE (bench_size <= 5.50 E shop_synergy_matches > 2.50 E gold > 0.50) ENTÃO [COMPRAR_UNIDADE_SINERGIA_OU_PAR] (confiança=93.2%, n=488)
  • SE (bench_size <= 5.50 E shop_synergy_matches <= 2.50 E lose_streak > 0.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=51.0%, n=412)

■ Fighting_Steel_Bastion_Specialist [STANDARD_ECON_50G] — Fidelidade da Árvore vs Rede Neural PyTorch: 78.8%
  Variáveis Decisivas: bench_size (61.5%), gold (19.3%), board_size (16.1%), shop_synergy_matches (1.7%), xp_to_next_level (1.4%)
  • SE (bench_size > 5.50 E board_size > 6.50 E gold > 3.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=74.1%, n=953)
  • SE (bench_size > 5.50 E board_size > 6.50 E gold <= 3.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=94.5%, n=441)
  • SE (bench_size <= 5.50 E gold > 1.50 E shop_synergy_matches > 1.50) ENTÃO [COMPRAR_UNIDADE_SINERGIA_OU_PAR] (confiança=96.5%, n=307)

■ Dark_Assassin_Crit_Specialist [HYPER_ROLL] — Fidelidade da Árvore vs Rede Neural PyTorch: 86.6%
  Variáveis Decisivas: bench_size (58.2%), relative_board_power (16.4%), gold (15.2%), win_streak (6.2%), shop_synergy_matches (4.0%)
  • SE (bench_size > 5.50 E relative_board_power <= 1.96 E win_streak <= 4.50) ENTÃO [OTIMIZAR_OU_VENDER_BANCO] (confiança=86.3%, n=2284)
  • SE (bench_size > 5.50 E relative_board_power > 1.96 E gold > 51.50) ENTÃO [REROLL_LOJA_1G] (confiança=65.4%, n=264)